UNIT- V STAFFING IN SOFTWARE PROJECTS

- Managing people
- Organizational behavior
- Best methods of staff selection
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- The Oldham
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Unit-V STAFFING IN SOFTWARE PROJECTS

- Whenever we develop a project, staff selection plays a vital role.
- Selection of suitable persons for the suitable job enables us to complete the project efficiently.
- Since a project includes various tasks, selection of staff for completing each task becomes essential.
- In this unit, we discuss how to manage people, how to select staff, how to motivate them in order to have a good and professional working environment.

MANAGING PEOPLE.

- is the process of managing those who work as individuals and in groups.
- People are considered to be the organization's most important assests.
- For a goal to be achieved effectively, in an organization, managing people is the most vital task.
- There are various activities involved in managing people. They are.

- Problem solving
- Motivating
- Planning
- Estimating
- Controlling and organizing.

• Problem solving:

- it is the process of dividing a problem in to sub-problems and assign each sub-problem to individual.
- mostly, problems are solved using available people i.e) the no. of staff who were working currently.

- Motivating:

  - it is the process of motivating the people who are currently working on a project.

  - This motivation process helps the people to develop positive attitudes and thereby yields better results.

  - There are various ways to motivate people. The management has to constantly motivate their staff members to produce a smooth working environment.

- Planning:

  - it deals with the planning of each and every task.

  - the duration of each and every task should be planned so that it can be finished accurately.

  - also planning ensures the people are working on the correct task with correct planning.

- Estimating:

  - this activity deals with the prediction of people about their performance.

  - this gives:
    - ✓ How fast people will work
    - ✓ How much time did they take to complete a task.
    - ✓ How fast they produce outputs.

- Controlling:

  - it is the process of controlling the activities involved in developing a project.

  - Controlling resources, controlling tasks assigned to each person form a vital task in this activity.

- Organizing:

  - it is the process of organizing the way in which people work.
  - it involves:
    - ✓ Preparing plan
    - ✓ Providing roadmap about a project
    - ✓ Organizing meetings to discuss about project
    - ✓ Providing alternative ways in case of any problem or defect.
    - ✓ Understanding people's behavior, etc.

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ORGANIZATIONAL BEHAVIOR.

- Studies about organizational behavior were done by Frederick Taylor in late 19th and early 20th Centuries.

- Taylor attempted to analyze the most productive way of doing manual tasks.

- The workers were then trained to do the work in this way.

- Taylor had 3 basic objectives:

- to select the best people for the job.
- to instruct them in the best methods
- to give incentives in the form of higher wages to the best workers.

- When developing a s/w project, the growth of both structured and agile methods should be taken into account.

- Taylor also emphasized on the importance of "performance related pay" (ie) the wages to the workers will be fixed depending on their performance.

Theories of organizational Behavior.

- In 1920, the researchers observed, when carrying out a famous set of tests on the conditions under which staff worked best.

- This research showed how the state of mind of workers influenced their productivity.

- This attitude was divided in to 2 types of theories. They are Theory x and Theory y.

Theory X: It holds that

- the average human has an innate dislike of work.
- there is a need therefore for coercion, direction and control
- people tend to avoid responsibility.

Theory Y: It holds that

- work is an natural as rest or play
- external control and coercion are not the only ways of bringing about effort directed towards an organization's ends
- commitment to objectives is a function of the rewards associated with their achievement.
- the average human can learn to accept and further seek responsibility
- the capacity to exercise imagination and other creative qualities is widly distributed.

Selecting the right person for the Job:

- this is also a part of the organizational behavior put forward by Taylor.
- He formulated the factor of selecting the right person for the right job.
- when we consider s/w development, many factors such as s/w tools and methods which affects productivity, should be taken in to account.
- The difference in the performance will be b/w the individual people, which is a vital factor.
- Hence, Taylor suggested that it is always better to have best people employed in the right work to get maximum productivity and getting good results.

Recruitment process:

- is a part of organizational behavior and responsibility.

- The project manager has the responsibility of choosing the right person to make up their team.

- Meredith Belbin categorizes people to recruitment process into 2 types. They are

i) Eligible Candidates
ii) Suitable Candidates

- Eligible candidates have a curriculum vitae (CV) which shows, for eg, the 'right' no. of years in some previous post and the 'right' paper qualifications.
- Suitable candidates can actually do the job well. While selecting a person for the job, actual skills of the person should be taken into account.

Recruitment process steps:

i) Create a job specification → Requirement of the job including the types of task to be carried out, should be agreed.

ii) Create a job holder profile → here, profile of the person should be constructed (ie) qualities, qualifications, education and experience of person.

iii) Obtain applications → Advertisement should be placed to the local press (ie) Newspaper, TV etc.
By giving the salary, location, job scope and other essential qualifications, the applicants will be limited to the more realistic candidates.

iv) Examine Curriculum Vitae → The applications should be read carefully and it is compared with job holder profile and suitable applications are filtered.

v) Conducting interviews → After examining curriculum vitae's of candidates, interviews are conducted.
- These include aptitude tests, personality tests and the examination of samples of previous work.
- major part of the interview would be evaluating and confirming statements in the CV.

vi) Other procedures → References will need to be taken up, wherever needed.

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Best Methods of staff selection

- The responsibility of the team leader becomes very high in case of recruiting new members into the team.

- This process is not easy, since the new members may be required in an organization, if the project is already in implementation stage.

- The team leader should be aware of these things and the newly recruited members should be assessed continually. This assessment helps to meet the demands of the project.

- Proper training should be given to each and every staff involved in the development of the project.

- The team manager should make decisions about whether the newly recruited staff member will get technical expertise on the project.

- Also, the team member should work on the project and try to learn about new tools and techniques.

- The training process must be actually implemented by the team members as a whole in order to meet the objectives.

- Reviews and inspections should be conducted regularly to get better results.

Motivation.

- Motivation is the term which means "encouragement".
- Encouraging the people to work is otherwise called as "motivation".
- motivating the people plays a vital role in developing a project, and thereby it produces good results.
- There are various theories formulated by different persons for motivating the people to work.
- Some of them are
  - Taylorist model
  - Maslow's hierarchy of needs
  - Herzberg's two-factor theory.
  - Expectancy theory of motivation.

Taylorist model:

- This model is reflected in the use of piece-rates in manufacturing industry.
- Piece-rates are those where workers are paid a fixed sum for each item they produce.
- These rates can cause difficulties if a new system will change work practices. Day-rates refer to payment for time worked.
- The amount paid to the workers will not directly relate to maximize the output in order to maximize their income.
- The amount of output will normally depend on working group and not based on an individual.
- Rewards based on piece-rates need to relate directly to work produced.

- As one member of staff in a study of s/w support work said: " This support department does well because we're a team, not because we're all individuals. I think it's the only way the support team can work successfully".

- In this kind of environment, a rewards system which makes excessive distinctions b/w co-employees could damage moral and productivity.

- In this method, bonus will be given to the product team members, when the project is successfully completed.

Maslow's hierarchy of needs:

- This theory is suggested by an American psychologist, Abraham Maslow. According to his principle, the motivation of an individual will vary.

- As a lower level of needs is satisfied then gradually a higher level of needs emerges.

- Maslow classified the needs into 2 types.

1. Basic needs → include food, shelter and personal safety.
2. High-level needs → is the need for " self-actualization". (ie) Complete fulfillment of potential.

- Normally, people gets motivated by different things at different stages of life.

- For eg, salary increases, while always welcome, probably have less impact on the more mature employee who is already relatively well paid than on a lowly paid trainee.

Herzberg's two-factor theory:

- Research into job satisfaction by Herzberg and his associates found 2 sets of factors about a job:

1. Hygiene or maintenance factors → which can make you dissatisfied if they are not right, for example the level of pay or the working conditions.
2. Motivators → which make you feel that the job is worthwhile, like a sense of achievement or the challenge of the work itself.

The expectancy theory of motivation:

- A model of motivation developed by Vroom and his colleagues.
- It identifies 3 influences on motivation.

1. Expectancy → the belief that working harder will lead to a better performance.
2. Instrumentality → the belief that better performance will be rewarded.
3. Perceived value → of the resulting reward.

- motivation will be high when all 3 factors are high. A zero level for any one of the factors can remove motivation.

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Oldham – Hackman Job Characteristics Model.

– This model is based on job satisfaction. Oldham and Hack man suggested that there are 5 factors which gives the job satisfaction.

– The first 3 factors make the job 'meaningful' to the person who is doing it:

- Skill variety → the no. of different skills that the job holder has the opportunity to exercise.
- Task Identity → the degree to which your work and its results are identifiable as belonging to you.
- Task Significance → the degree to which your job has an influence on others.

– The other two factors are:

- Autonomy → the discretion you have about the way that you do the job.
- Feedback → the information you get back about the results of your work.

Methods of improving motivation:

– To improve motivation the manager might therefore do the following:

- Set specific goals → These goals need to be demanding and yet acceptable to staff. Involving staff in the setting of goals helps to gain acceptance for them.
- Provide feedback → not only do goals have to be set but staff need regular feedback about how they are progressing.

- Consider job design → Jobs can be altered to make them more interesting and give staff more feeling of responsibility.

  – 2 measures are often used to enhance job design.

- Job enlargement → The person doing the job carries out a wider variety of activities. It is the opposite of increasing specialization.
- Job enrichment → The job holder carries out tasks that are normally done at a managerial or supervisory level. With programmers in a maintenance team, they might be given authority to accept requests for changes that involve less than 5 days work without the need for their manager's approval.

Ethical and programmed Concerns.

- Ethical and Professional responsibilities have a major role to play in every Project management.

- Some ethical responsibilities are shared by all members of the community irrespective of their position. For eg, if an accident takes place in an organization the emergency services should be alerted by anybody to escape from the accident.

- However, some roles and responsibilities affect the particular people. (ie) person of specific expertise, for example, Consider a s/w engineer(or) a IT practitioner.

- Let us consider a s/w engineer and have a look at his ethical responsibilities code of ethics for a s/w engineer. This code of ethics was developed by ACM/IEEE-CS joint task force on s/w engineering ethics and professional practices.

- There are 8 principles involved in this code of ethics.

1. Public:

s/w engineer shall act consistently with the public interest.

2. Client and employer:

s/w engineer shall act in a manner that is in the best interests of their client and employer consistent with the public interest.

3. product:

s/w engineers shall ensure that their products and its related modifications meet the highest professional standards as possible.

4. Judgement:

S/w engineer shall maintain integrity and independence in their professional judgement.

5. Management:

S/w engineering managers and leaders shall subscribe to promote an ethical approach to the management of s/w development and maintenance.

6. Profession:

S/w engineers shall advance the integrity and reputation of the profession consistent with the public interest.

7. Colleagues:

S/w engineers shall be fair to and supportive of their colleagues.

8. Self:

- S/w engineers shall participate in continuous learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

- Hence, these principles are the code of conduct put forward by ACM for s/w engineers.

- Some professions such as medicine have their own ways of certifying the practitioners by enforcing ethical codes of conduct.

- BCS (British Computer Society) in UK has produced codes of conduct and good practices.

- ACM has produced code of conduct in the US.

- Thus the organization should have a mechanism to regulate this ethical concerns and this should be done by a manager.

- The manager should evaluate these issues and take necessary actions.

- This also includes taking this issues to higher level of management and providing solutions.

Working in Teams.

The main objectives of the teams such as

i) Improve group working
ii) Analyze the coordination of a project
iii) Select the best communication genres to support the Coordination needs of a project.
iv) To evaluate the characteristics of the various team structures.

Introduction:

- S/w-based systems can be huge the s/w to control a telephone switching system can contain 5 million of code.
- It will look enhancing communication b/w individual developers within teams and across teams.
- By the 'teams', usually mean groups of people who working. Typically the individuals work in the same office, that is co-located although it is a same.
- The team 'project_team' is sometimes used to refer to all the people working on a project. Thus the individual s/w developers are likely to transfer b/w teams as projects start and finish.
- A team is carried out a joint assessment. An organization needs to control the allocation of staff to work assignments.
- The collaborative nature of project work will have an influence on nearly all stages of stepwise project planning framework.

1. Identify project scope and objectives
2. Identify project infrastructure
3. Analyze project characteristics
4. Estimate for each effort activity
5. Identify activity risk.
6. Allocate resources
7. Review) publicize plan.

Becoming a Team:

To become a team process we categorize 5 basic stages of development.

i) forming → The members of the group get to know each other and try to set up some ground rules about behaviour.

ii) Storming → Conflicts arise as various members of the group try to exert leadership and the group's methods of operation are being established.

iii) Norming → Conflicts are largely settled and a feeling of group identity emerges.

iv) Performing → The emphasis is now on the tasks at hand.

v) Adjourning → The group disbands.

Valuable research has examined the best mix of personalities in a project team. Belbin came to the conclusion teams needed a balance of different types of people:

1) The chair → not necessarily brilliant leaders but they must be good at running meetings, being calm, strong but tolerant.

2) The Plant → Someone who is essentially very good at generating ideas and potential solutions to problems.

3) The monitor - evaluator → good at evaluating ideas and potential solutions and helping to select the best one.

4) The shaper → rather a worrier, who helps to direct the team's attention to the important issues.

5) The team worker → skilled at creating a good working environment.

6. The resource investigator → adept at finding resources in terms of both physical resources and information.

7. The completer-finisher → Concerned with completing tasks.

8. The company worker → a good team player who is willing to undertake less attractive tasks if they are needed for team success.

Group performance:

- It is better than individuals work. The process is said to be some work yields better results if carried out as a team while some things are slowed down if the work is compartmentalized on an individual basis.

- One way categorizing group tasks is into,

i) Additive tasks → mean that efforts of each participant are added to get the final result.
(eg) Gang clearing snow.

ii) The compensatory tasks → the judgements of individual group members are pooled so that the errors of some are compensated for by the inputs from others.
(eg) Individual members of a group are asked to provide estimates of effort needed to produce a piece of s/w and results are then averaged.

iii) Disjunctive tasks → there is only one correct answer. The effectiveness of group depends on, someone coming up with the right answer.

iv) Conjunctive tasks → are where progress is governed by the rate of slowest performer.
(eg) s/w production where progress is governed by the rate of the slowest performer.

Decision Making

- Before we can look more closely at the effectiveness with which groups can make decisions, we need to look in general terms at the decision-making process.

- Decisions can be categorized as being:

- Structured → generally relatively simple, routine decisions where rules can be applied in a fairly straightforward way, or
- Unstructured → more complex and often requiring a degree of creativity.

- Another way of categorizing decisions is by the amount of risk and uncertainty that is involved.

Some mental obstacles to Good Decision Making:

- To access the rightly stressed a structured, rational, approach to decision making. Many management decisions in the real world however are made under pressure and based on incomplete information.

- We aware of mental obstacles to effective intuitive thinking. For example,

i) Faulty Heuristics – Heuristics or 'rules of thumb' can be useful but there are dangers,
   - They are based on information that is to hand, which misleading.
   - They are based on steady per, such as accepting a welshmen into a male voice choice without an audition because of "well known" fact the welsh are a great singing nation.

ii) Escalation of Commitment – This refers to the way that once you have made a decision it is increasingly difficult to alter it even in the face of evidence is wrong.

iii) Information overload – It is possible to have too much information.

Group Decision Making:

- Decision made by the team as a whole are more likely to be accepted than those that are imposed.

- Assume that the meetings are genuinely collectively responsible and have been properly briefed, research would seem to show that groups are better at solving complex problems when the members of the group have complementary skills and expertise.

- Groups deal less effectively with poorly structured problems needing creative solutions.

- Brainstorming techniques can help groups in this situation but research shows that people often come up with more ideas individually than in a group.

Obstacles to Good Decision making:

Amanda could find that group decision making have disadvantages.

- It is time consuming
- It is stirrup conflicts within group.
- Decisions can be unduly influenced by dominant personalities.
- Conflict can, in fact be less than might be expected.

Measures to Reduce the disadvantages of Group decision making.

- One method of making group decision making more efficient and effective is by training members to follow a set of procedure

- Delphi technique endeavours to collate the judgements of a no. of experts without actually bringing them face to face. Given a problem the following procedure is carried out,

- ✓ The cooperation of a no. of experts is enlisted.
- ✓ The problem is presented to experts.
- ✓ The experts record their recommendations.
- ✓ The collected responses are recirculated.
- ✓ The experts comment on the ideas of others and modify their recommendations if so moved.

Team Headfulness:

A successful move can be triggered where one players sees that another is in position to score a goal if provided with a ball. This is an example of team headfulness, where group members are aware of the activities of others to contribute to overall group success and can identify ways of supporting that contribution.

Egoless Programming:

- The earlier days of computer development managers tended to think of s/w developer as communing mysteriously with the machine.

- The tendency was for the programmers to see programmers as being an extension of themselves and to feel over protective towards them. The effects of this on the maintainability of programs can be imagined.

- Programs would become in effect of common property of the programming group and programming would become 'egoless'.

Chief Programmer Teams:

- The larger the development group the slower it becomes because of the increased communication. Thus large time-critical projects tends to have a more formalized, centralized structure.

- The result was the chief programmer team. He/she defines the specification and designs, codes, tests and documents the S/W. He or she is assisted by a co-pilot with whom the chief programmer can discuss problems and who writes the code.

Extreme Programming (XP):

- The new XP concepts have inherited some of these ideas. Most XP practices can be seen as ways of promoting a collective mind.

- The advantages of XP argue this is a self-defeating. They suggest other, less formal methods of communication and coordination.

- The scrum method is used in the process of extreme programming.

Scrum:

- Scrum S/W development process illustrates some of the points as it has many elements found in agile methods but also has an element of chief programmer philosophy.

- Scrum process starts with a systems architecture and planning phase. This has something of the programmer's of chief approach as a chief architect defines the overall architecture of the product.

- This phase is now followed by a no. of sprints, each of them which typically last b/w one or 4 weeks. It is possible for scrum teams to work in parallel on diff. prints, but all teams must finish their sprint on the same way.

- Sprints are time-boxed and at the end of the sprint period some uncompleted, lower priority features may be held over.

- When all the sprints have been completed, there is a final closure phase where tasks like regression and integration testing and the writing of user and training guides take place to create a final package for delivery to market.

Team Structures.

Organization and Team structures:

– Large s/w development companies are usually organized into departments. Departmentalization of a company may be based on several criteria such as staff specialization, product lines, categories of customers, or geographical location.

– 2 important issues that are critical to the effective functioning of every organization are:

- Department structure → How is a department organized into teams?
- Team structure → How are project team structured?

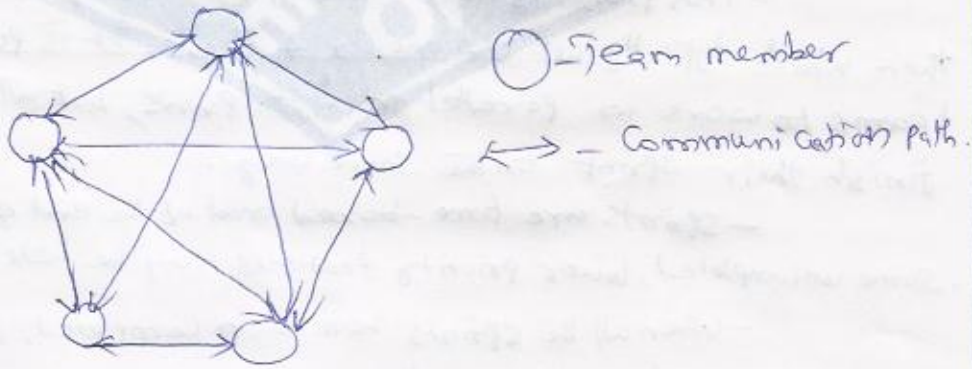
Team structure:

It denotes the reporting, responsibility and communication structures in individual project teams. We consider only 3 team structures.

1. Democratic
2. Chief programmer
3. Mixed team organizations.

1) Democratic team:

This structure, as the name implies, does not enforce any formal team hierarchy. Decisions are taken based on discussions, where any member is free to discuss with any other member as shown below.

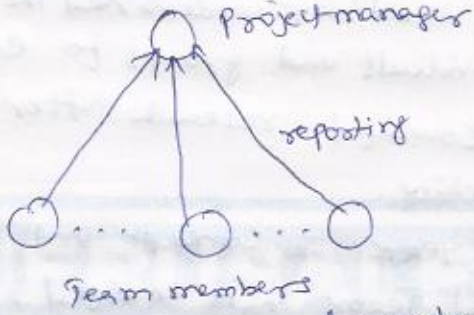


Democratic team structure.

– A manager provides the administrative leadership.

2) Chief Programmer team:

- A schematic representation of the chief programmer team structure has been shown below.

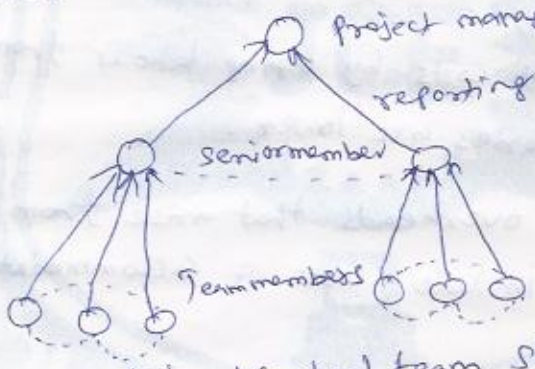


Chief Programmer team structure.

- In this team structure, a senior member provides the technical leadership and is designated as the chief programmer.

3) Mixed Control team structure:

- It draws ideas from both the democratic and chief-programmer team structures.



Mixed Control team structure

- the communication paths are shown as dashed lines and reporting structure is shown using solid arrows.

Virtual Teams

- The heart of many projects is collaborative problem solving. The second world war underlined the importance of cooperation b/w individuals and groups to execute major global operations and encouraged research after the conflict into effective team working.

- The recent years the concept and practice of having dispersed or "Virtual Teams" have emerged.

- The nature of the work carried out in some of the projects means that the demand for certain specialist skills is intermittent.

- It is then only a step to use "off-shore" staff who live and work in a different part of world. To recap the possible advantages of such an arrangement,

i) A reduction in staff costs by using labour from develop the countries where salaries are lower.

ii) A reduction in the overheads that arise from having the own employees on site, include costs of accommodation social security payments training.

iii) The flexible use of staff - they aren't employed when they are not needed.

iv) Productivity might be higher.

v) Use of specialist staff for specific jobs rather than more general project workers might improve quality.

vi) Advantages of these technique, it can be taken of people working in different time zones to reduce task durations.

- For example s/w developers can deliver new versions of code to testers in a different time zone who can test it and deliver the results back at the start of the next working day.

- Some of the challenges of dispersed working are,

1) The requirements for work that is distributed to contractors have to be carefully specified.
2) Coordination of dispersed workers can be difficult.
3) There may be a lack of trust of co-workers who are remote and never seen.
4) Assessment of the quality of delivered products will need to be thorough.
5) Different time zones can cause communications and coordination problems.

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Communication Genres.

- are used work groups but with colleagues in other parts of the organization, or some cases in partner organizations.

- Some of the work has been done to study the characteristics of various methods of communication one approach has been to try identify communication genres. These are refer to the technical means of communications.

- For some types of communication such as official meetings these rules might be quite formal.

- With in the general heading of "management meetings" there could be seen as genres in their own right.

- The general heading of e-mail communication genre, advanced email based application can be developed where the content of the email is structured by preset proformas designed to fulfil a standard process, such as requesting change to a s/w specification.

- A major influence on the nature of communication genres is the constraints of time and place.

- modes of communication can be categorized as combinations of 2 opposites: same time/different time and same place/different place described below.

	Same place	Different place
Same Time	meetings, interview, etc	Telephone, Instant message, etc.
Different Time	Notice board, pigeon holes	Email, Voicemail, Documents.

- The nature of the data to be conveyed also needs to be considered.

1) What is the extent and complexity of the data to be conveyed?

A telephone conversation may be a very quick way of convey simple message.

2) Is it easy to understand?

for example is the context well known to both sender and receipient? if it is likely that receipient will need clarification of aspects of the data, then a mode of communication is 2 way.

3) Where the communication is personally sensitive, then face-to-face contact is the most effective form of communication even if it can be uncomfortable or inconvenient for those concerned.

A diff. stages of project it is likely that different communication genres will be preferred.

Early stages of project:

- At the start of the project, team members will need to build up their trust and confidence of their co-workers.

- The dispersed projects as Charles handy has written, "paradoxically the more virtual an organization becomes, the more its people need to meet in person".

- The early stages of a project are when at least some of the team will be involved in making decisions about the overall design of the product to be delivered and method by which it is to be created.

- In the preliminary stage at least same time/same place meetings are the most effective means of progressing.

Intermediate design stage of project:

- Once the overall structure of the architecture of product has been displayed detailed design on different components well be carried out in parellel in various locations.

- Few of them points need to be clarified and for this, same time/different place communication could well be best such as the use of teleconferencing.

Implementation stage of the project:

- once the design is verified and everyone knows his or her role work can progress.

- Where there is a need to exchange data at this point, various time/various place communication media such as email are likely to be sufficient.

- Even the process of the stage is regular face-to-face meetings of at least some staff as this supply a rhythm to the project which helps coordination of the project and maintain motivation.

- The dispersed project such as, "Interaction b/w coordination meetings was mainly in response to the previous meeting or in anticipation of the next one. The coordination meeting served as a heartbeat rhythmically new life into team process".

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Communication Plans.

- Communication is the main for all the projects but a vital matter of dispersed projects. The consideration of project stack-holders will communicate ought be a part of project planning process.

- To suggest specific planning project document ought to address communication issues affecting the project.

- The first step produce a plan is to list the main stakeholders, with spatial attention to those who are participate in the development and implementation of project - it may be reuse identification of stackholders.

- It required different phases of a dispersed project. Consultation with the representatives of various groups of stackholders will be essential part of this process.

- The results of this process will be documented in a table for the following column headings.

What - This contains name of particular communication event.

eg. i) Kick-off meeting
ii) Project intranet site.

- Who) target the audience for the communication. Target audience mayn't convey quite the right idea as this implies that there are passive recipients of data from a central authority.

Purpose – What the Communication is to achieve.

When) frequency – If the communication by single event, then a specific date can be supplied.

If the event is recursing one such as progress meeting then the frequency should be indicated.

- ✓ Type/method – Nature of the communication.
  - ✓ For example a meeting or a distributed document.
  - ✓ Responsibility – The person who initiates the communication.

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